Wentao Zhang

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Work Experience

University of Nottingham 2025/09 - Now

Research Assistant Nottingham, United Kingdom

- Power Electronics and Machines Centre (PEMC)
- Developing in-house finite element analysis code package for electromagnetic, structural modal, and mechanical stress analyses of electrical machines
- Extending operator-learning neural surrogates (deep learning-based magnetic field models) to replace expensive FEA evaluations, enabling rapid design-space exploration and inverse design
- Applied to next-generation magnet-free synchronous machine development

Southeast University / Inovance 2021/09 - 2025/09

EV Traction Machine R&D (PhD research in collaboration with industry) Nanjing / Suzhou, China

- Collaborated with Inovance, a world-leading EV traction machine supplier, to design, prototype, and test two advanced oil-cooling PM traction machines through full industrial development cycles
- **20,000 rpm PM traction machine for A-class vehicle:** measured peak power 270 kW, peak torque 380 Nm, power density 7.87 kW/kg, max efficiency 97.4%
- **18,000 rpm PM traction machine for A0-class vehicle:** measured peak power 113 kW, peak torque 138 Nm, power density 7.78 kW/kg, max efficiency 95.3%
- Design stage: built a multi-physics optimisation framework integrating electromagnetic FEA, lumped-parameter thermal network modelling, and mechanical stress evaluation
- Manufacturing stage: produced engineering drawings for electromagnetic and structural components, gained experience in tolerance specification for high-reliability manufacturing
- Testing stage: characterised both prototypes on an AVL dynamometer, extracting efficiency maps, key performance indicators, and voltage/current waveforms

Education

Southeast University 2021/09 - 2025/09

Ph. D. in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Multi-physics Analysis and Design of Hairpin Winding PM Traction Machines in Electric Vehicles"
- 2025 1st Prize, IEEE TEC Electric Vehicle Design Competition
- 2025 2nd Prize Paper Award, IEEE Transactions on Transportation Electrification

Southeast University 2018/09 - 2021/06

Master in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Analysis and Design of High-Performance Wound-Excited Magnetic Field Modulation-based Brushless Machines"

Southeast University 2014/09 - 2018/06

B.Eng in Electrical Engineering Nanjing, Jiangsu, China

- Dissertation Title: "Comparative Study of Control Systems for PM Machine in Electric Vehicles"

Technical Expertise

My core expertise is the multi-physics design, optimisation, and testing of high-power-density electric traction machines for EVs, augmented by AI-driven surrogate modelling. I have taken two PM traction machines from concept through prototyping to dynamometer validation under full industrial development cycles, achieving power densities up to 7.87 kW/kg and peak efficiencies exceeding 97%. I have also developed an in-house FEA code package (electromagnetic, structural modal, and mechanical stress analyses) and operator-learning neural surrogates that replace expensive FEA evaluations, now applied to next-generation magnet-free synchronous machine development at Nottingham. My expertise spans five areas.

(1) Multi-physics motor design and optimisation. Electromagnetic FEA, lumped-parameter thermal network modelling, and mechanical stress evaluation are coupled within a multi-objective optimisation framework. I have applied this to PM machines, wound-field switched flux machines, and hybrid excitation topologies, balancing torque density, efficiency, flux regulation capability, and manufacturability constraints. **(2) Hairpin winding and advanced winding technology.** I developed a systematic n-layer flat wire winding design methodology for high-power-density EV PM machines, covering conductor layout optimisation, AC loss reduction, and experimental validation on prototype hardware. **(3) Thermal management and cooling system design.** I have built high-fidelity lumped-parameter thermal networks and conducted fluid-solid coupled oil-cooling analysis for hairpin-winding PM machines, validated against direct thermocouple measurements on dynamometer-tested prototypes. **(4) Power electronics and machine control.** My work addresses PWM excitation effects on machine losses and temperature, spatial harmonic modelling of machine inductance, and converter-machine interaction analysis. I have also designed and implemented a three-phase full-bridge converter and associated control algorithm for testing flux-modulation machine prototypes. **(5) AI for science and learned surrogate modelling.** I have developed a deep-learning magnetic-field surrogate (ICEM 2026, accepted) using operator-learning neural networks that replace finite-element evaluations with fast learned operators, enabling rapid forward modelling, design-space exploration, and inverse design at orders-of-magnitude reduced computational cost. This methodology is directly applicable to accelerating the multi-objective optimisation workflows central to next-generation traction machine development.

Research Publication

Transaction/Journal Paper (22 papers, 17 in IEEE Transactions)

[J1] **Wentao Zhang**, W. Hua, Z. Wu, G. Zhao, Y. Wang, and W. Xia, "Analysis of DC Winding Induced Voltage in Wound-Field Flux-Switching Machine With Air-Gap Field Modulation Principle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2300–2311, Mar. 2022.

[J2] **Wentao Zhang**, Z. Wu, W. Hua, Z. Zhu, Z. Wu, J. Hu, et al., "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth Tips," *IEEE Trans. Ind. Appl.*, vol. 58, no. 2, pp. 1976–1990, Mar.–Apr. 2022.

[J3] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," *IEEE Trans. Ind. Appl.*, vol. 59, no. 4, pp. 3966–3977, Jul.–Aug. 2023.

[J4] **Wentao Zhang**, Z. Wu, Y. Fan, W. Hua, and M. Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 60, no. 4, pp. 6023–6032, Jul.–Aug. 2024.

[J5] **Wentao Zhang**, Z. Wu and W. Hua, "Design of n-Layer Flat Wire Winding in Permanent Magnet Machine for Electric Vehicles," *IEEE Trans. Transport. Electrific.*, vol. 11, no. 3, pp. 7798–7810, 2025.

[J6] **Wentao Zhang**, Z. Wu, L. Jin, Y. Fan, W. Hua, and M. Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2996–3006, Mar.–Apr. 2025.

[J7] **Wentao Zhang**, Z. Wu, Z. Wang, W. Hua, Y. Fan, Z. Xu, et al., "Design Tradeoff Between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," *IEEE Trans. Ind. Appl.*, vol. 61, no. 5, pp. 7198–7208, Sep. 2025.

[J8] **Wentao Zhang**, Y. Fan, Z. Zhu, Z. Wu, W. Hua, and M. Cheng, "Analysis of DC Winding Induced Voltage in Wound-Rotor Synchronous Machines by Using the Air-Gap Field Modulation Principle," *World Electr. Veh. J.*, vol. 13, no. 11, Art. no. 215, 2022.

[J9] **Wentao Zhang**, Z. Wu, Z. Cang, W. Hua, Y. Fan, and M. Cheng, "Influence of End-Effect in Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 62, no. 1, pp. 785–798, Jan. 2026.

[J10] **Wentao Zhang**, Z. Wu and W. Hua, "High-Fidelity Lumped Parameter Thermal Model for Oil-Cooling Hairpin Winding PM Machines in EVs," *IEEE Trans. Transport. Electrific.*, vol. 12, no. 2, pp. 2978–2991, Apr. 2026.

[J11] Z. Wu, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "A Transient-State Lumped Parameter Thermal Model for Brushless Wound Field Switched Flux Machines," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 1, pp. 224–238, Mar. 2024.

[J12] Z. Wu, Z. Zhu, C. Wang, W. Hua, K. Wang, and **Wentao Zhang**, "Influence of Rotor Iron Bridge Position on DC-Winding-Induced Voltage in Wound Field Switched Flux Machine Having Partitioned Stators," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 20–28, Sep. 2021.

[J13] Z. Wu, Z. Zhu, S. Cai, W. Hua, and **Wentao Zhang**, "Enhancement of Torque Density in Wound Field Switched Flux Machines With Partitioned Stators Using Assisted Ferrites," *Chin. J. Electr. Eng.*, vol. 7, no. 3, pp. 42–51, Sep. 2021.

[J14] Z. Wu, L. Jin, W. Hua, **Wentao Zhang**, and M. Cheng, "Harmonics Orders Modeling of DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines Under PWM Excitation," *IEEE Trans. Transport. Electrific.*, vol. 10, no. 3, pp. 7040–7050, Sep. 2024.

[J15] Z. Wu, L. Jin, **Wentao Zhang**, Y. Fan, W. Hua, and M. Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage Pulsation in Wound Field Switched Flux Machines," *IEEE Trans. Ind. Appl.*, vol. 60, no. 1, pp. 460–476, Jan.–Feb. 2024.

[J16] Z. Wu, Z. Zhu, W. Hua, S. Akehurst, X. Zhu, **Wentao Zhang**, et al., "Analysis and Suppression of Induced Voltage Pulsation in DC Winding of Five-Phase Wound-Field Switched Flux Machines," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 1890–1905, Dec. 2019.

[J17] L. Jin, W. Hua, U. B. Akuru, Z. Wu, **Wentao Zhang**, and M. Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," *IEEE Trans. Ind. Appl.*, vol. 61, no. 2, pp. 2874–2883, Mar.–Apr. 2025.

[J18] X. Yan, Z. Wu, W. Hua, **Wentao Zhang**, and M. Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6811–6822, Sep.–Oct. 2024.

[J19] H. Hua, X. Chen, D. Li, **Wentao Zhang**, Z. Wu, and W. Hua, "Investigation of Torque Improvement of Modular Stator Surface-Mounted Permanent Magnet Machines With Flux Gaps," *IEEE Trans. Ind. Appl.*, vol. 60, no. 5, pp. 6787–6798, Sep.–Oct. 2024.

[J20] J. Chen, W. Hua, L. Shao, Z. Wu, X. Yan, and **Wentao Zhang**, "Modified Magnetic Equivalent Circuit of Double-Stator Single-Rotor Axial Flux Permanent Magnet Machine Considering Stator Radial-End Flux-Leakage," *IET Electr. Power Appl.*, vol. 18, no. 2, pp. 195–207, Feb. 2024.

[J21] Z. Wu, X. Yan, Z. Wang, **Wentao Zhang**, and W. Hua, "Influence of PWM Excitation on Loss and Temperature of Flat Wire Permanent Magnet Machines for Electric Vehicles," *IEEE Trans. Ind. Electron.*, vol. 73, no. 4, pp. 5174–5185, Apr. 2026.

[J22] Z. Wu, E. Fu, J. Du, **Wentao Zhang**, and W. Hua, "Quantitative Comparison of Electromagnetic Vibration of PMSMs With Glued and Welded Stator Core for Electric Vehicles," *IEEE Trans. Ind. Appl.*, early access, 2025.

Conference Paper (18 papers)

[C1] **Wentao Zhang**, Zhongze Wu and Wei Hua, "A Deep Learning Based Magnetic Field Surrogate Modeling Method for Interior Permanent Magnet Machine in EVs," in Proc. Int. Conf. Elect. Mach. (ICEM), Funchal, Madeira, Portugal, 2026. (**Accepted**)

[C2] **Wentao Zhang**, Zhongze Wu, Wei Hua and Ziqiang Zhu, "Reduction of Open-Circuit DC Winding Induced Voltage and Torque Pulsation in the Wound Field Switched Flux Machine by Stator Axial Pairing of Tooth-Tips," in Proc. IEEE Int. Elect. Mach. Drives Conf., Gothenburg, Sweden, 2020, pp. 522-528.

[C3] **Wentao Zhang**, Zhongze Wu, Ying Fan, Wei Hua and Ming Cheng, "A Stable and Computationally Efficient Spatial Harmonic Model for Predicting the DC Winding Induced Voltage in WFSF Machine," in Proc. IEEE Energy Convers. Congr. Expo., Detroit, MI, USA, 2022, pp. 1-7.

[C4] **Wentao Zhang**, Zhongze Wu, Wei Hua, Ying Fan, Zhiyuan Xu and Ming Cheng, "Design Tradeoff between Flux Regulation Capability and DC Winding Induced Voltage in Hybrid Excitation Switched Flux Machine," in Proc. IEEE Energy Convers. Congr. Expo., Nashville, TN, USA, 2023, pp. 5060-5066.

[C5] **Wentao Zhang**, Zhongze Wu, Lai Jin, Ying Fan, Wei Hua and Ming Cheng, "Analysis and Multi-Objective Optimization of the Hybrid Excitation Switched Flux Machine," in Proc. Int. Elect. Energy Conf., Hefei, China, 2023, pp. 3371-3376. (**Best Paper Award**)

[C6] **Wentao Zhang**, Zhongze Wu, Wei Hua, Ying Fan and Ming Cheng, "Influence of the End-Effect in Wound Field Switched Flux Machine," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 1223-1227.

[C7] **Wentao Zhang**, Zhongze Wu, Ying Fan, Wei Hua and Ming Cheng, "Analysis and Improvement of DC Winding Current Dynamics in Wound Field Switched Flux Machine," in Proc. Int. Conf. Energy Technol. Future Grids, Wollongong, Australia, 2023, pp. 1-6.

[C8] **Wentao Zhang**, Zhongze Wu, Wei Hua and Ji Qi, "Analysis of PM Volume Reduction Methods for Hybrid Excitation Switched Flux Machines," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2024, pp. 3152-3157.

[C9] **Wentao Zhang**, Zhongze Wu and Wei Hua, "Field-Oriented Design of Notching and Mechanical Stress Relieving Grooves for Rotors in Flat Wire Permanent Magnet Machines," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 304-309.

[C10] Zhongze Wu, Lai Jin, **Wentao Zhang**, Ying Fan, Wei Hua and Ming Cheng, "Influence of PWM Excitation on DC Winding Induced Voltage in Wound Field Switched Flux Machines," in Proc. IEEE Energy Convers. Congr. Expo., Detroit, MI, USA, 2022, pp. 1-7.

[C11] Lai Jin, Wei Hua, Udochukwu B. Akuru, Zhongze Wu, **Wentao Zhang** and Ming Cheng, "Comparative Study on DC Winding Induced Voltage Pulsation of Wound Field Flux Modulation Machines Having Different Iron Core Structures," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 442-447.

[C12] Martin Philip Koroma, Wei Hua, Zhongze Wu, Lai Jin and **Wentao Zhang**, "Influence of Magnet Layer Numbers on Electromagnetic Performance of Interior Permanent Magnet Machines," in Proc. Int. Conf. Elect. Mach. Syst., Zhuhai, China, 2023, pp. 696-700.

[C13] Xinying Chen, Hao Hua, Dalin Li, **Wentao Zhang**, Zhongze Wu and Wei Hua, "Modular Stator Surface-Mounted Permanent Magnet Machines with Flux Gaps," in Proc. IEEE Int. Elect. Mach. Drives Conf., San Francisco, CA, USA, 2023, pp. 1-7.

[C14] Xueyi Yan, Zhongze Wu, Wei Hua, **Wentao Zhang** and Ming Cheng, "A Mathematical Model for Wound Field Switched Flux Machine Considering Inductance Harmonics of Field and Armature Windings," in Proc. IEEE Int. Elect. Mach. Drives Conf., San Francisco, CA, USA, 2023, pp. 1-6.

[C15] Zhongze Wu, Lai Jin, **Wentao Zhang**, Xueyi Yan, Haorui Ge, Hao Hua, Wei Hua and Ming Cheng, "A Lumped Parameter Thermal Model for Variable Flux Reluctance Machine Considering Relative Position of DC and AC Windings," in Proc. Int. Conf. Elect. Mach., Torino, Italy, 2024, pp. 1-7.

[C16] Zhongze Wu, Xueyi Yan, Zhimian Wang, **Wentao Zhang** and Wei Hua, "Quantitative Analysis of the Torque-Speed Curve for a Flat Wire Permanent Magnet Machine Drive System," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 40-44.

[C17] Zhongze Wu, Erfang Fu, Jinwen Du, **Wentao Zhang** and Wei Hua, "Influence of Stator Core on the Vibration Behavior of Permanent Magnet Machines with Hairpin Winding," in Proc. IEEE Int. Elect. Mach. Drives Conf., 2025, pp. 34-39.

[C18] Zhongze Wu, **Wentao Zhang**, Zhimian Wang, Lingyun Shao, Wei Hua and Ming Cheng, "Stator Semi-Skewing Method for High-Speed Permanent Magnet Machine," in Proc. Int. Conf. Elect. Mach. Syst., 2025, pp. 693-697.

Patent

- Zhongze Wu,**Wentao Zhang**, Lai Jin, and Wei Hua, "Inductive Ripple Voltage Suppression Method Based on Low-Pass Filter Circuit," CN118508832A, 2024.
- Zhongze Wu, **Wentao Zhang**, Lai Jin, and Wei Hua, "Disk Brushless Electric Excitation Magnetic Field Modulation Motor With Double-Row Stator Structure," CN118100572A, 2024.
- Zhongze Wu, **Wentao Zhang**, Huilin Huang, and Wei Hua, "Method for Constructing a High-Precision Motor Performance Database Based on a Dual-Model Strategy," CN117972924A, 2024.
- Zhongze Wu, **Wentao Zhang**, Wei Hua, and Zhixiang Zou, "Method for Establishing General Lumped Parameter Thermal Network Model of Motor," CN117094088A, 2023.
- Zhongze Wu, **Wentao Zhang**, Lai Jin, and Wei Hua, "Method for Analyzing Harmonic Orders of Induced Ripple Voltage of Excitation Winding," CN115642847A, 2023.
- Zhongze Wu, **Wentao Zhang**, Junxiang Liao, Ying Fan, Wei Hua, and Ming Cheng, "Excitation Winding Induced Pulse Voltage Suppression Method Based on Harmonic Current Injection," CN115208245A, 2022.
- Zhongze Wu, **Wentao Zhang**, Ying Fan, Wei Hua, and Ming Cheng, "High-Precision Field Path Coupling Design Method for Brushless Electric Excitation Magnetic Field Modulation Motor," CN114781227A, 2022.
- Zhongze Wu, **Wentao Zhang**, Wei Hua, and Ziqiang Zhu, "Method for Weakening the Induced Pulsating Voltage of a Brushless Electric Excitation Magnetic Field Modulated Motor," CN114189185A, 2022.
- Wei Hua, **Wentao Zhang**, Zhongze Wu, and Shichuan Ding, "Method for Reducing Torque Ripple and Induced Voltage of Electric Excitation Flux Switching Motor," CN111769697A, 2020.

Awards

- 2025 1st Prize, IEEE TEC Electric Vehicle Design Competition
- 2025 2nd Prize Paper Award, IEEE Transactions on Transportation Electrification
- 2024 2nd Prize in 11th Energy Equipment Innovative Design Competition for China Postgraduate
- 2023 **Wiscom System Scholarship**
- 2023 1st Prize Academic Scholarship
- 2023 CIEEC Best Paper Award

- 2023 **1st Prize in IEEE IAS CMD Thesis Contest (non-Ph. D)**
- 2022 Outstanding Master Degree Thesis in Jiangsu Province
- 2017 2nd Prize in Southeast University Smart Car Competition

Last Updated on May 14, 2026